

BrickAtlas: A Benchmark for Visual Evidence Following

Supplementary Material

Anonymous CVPR submission

Paper ID Draft

1. Resource and Observation Inventory

This supplement documents the source collection, exact visual observations, assignment audits, qualification protocol and supporting graph controls. It accompanies the current benchmark manuscript. The primary visual dataset remains `brickatlas-display-v3`; subsequent measurement audits do not create a new stimulus version. The 15-model experiment design is a proposal, independent of the sealed earlier three-model study.

The 24 human-authored LDraw sources contain 15,334 placed parts. Source URLs, credits, licenses, original files and hashes are retained. The 617 historical dossiers preserve questions and inspection actions. Only 140 parents contribute primary visual pairs: 73 Color and 67 Part-type. Table 2 lists their source coverage. Table 1 groups the historical resource by assembly scale. The scale bands are not validated cognitive-difficulty levels. Figure 3 presents every source using the original full-assembly renders, with source ID, scale band and part count. Together with the main-paper examples, it separates the physical source collection from its task-specific observations.

Table 1. Source scale and historical item counts. These item counts must not be interpreted as primary paired observations.

Source band	Sources	Parts	Items
D1	5	501	83
D2	9	2,172	175
D3	2	1,440	59
D4	8	11,221	300

1.1. Exact input boundary

An observation binds the complete system and user messages, question, options and native 1280×800 image. The internal record retains its observation ID, source identity, display mapping, condition and gold. The model-visible input does not expose these audit fields. All labels resolve inside the current image. The prescribed output is exactly

one JSON object containing one listed `choiceId`. The exact shared identity contract is:

References select objects in the supplied image only. A label such as B0001 is an image-local display reference, not a request to retrieve a historical source instance. Renders may have edited appearances or label assignments. Answer only from the current image.

Changes to text, options, image bytes, dimensions or image policy require renewed qualification of affected observations. Provider envelopes additionally record requested model revision, adapter, generation settings and full request hash. The provider’s internal image preprocessing is not directly observed. A matching local envelope therefore establishes the delivered request, not every internal transformation.

1.2. Concrete visual pair

For OMR 42004, Color asks for the current color of B0035. The displayed answer is Blue in A and Red in B; the original source record stores Black. The image-local contract explicitly permits edited appearance. Part-type retains B0035 as target. Candidate B0036 selects the matching source type, 32250, in A. After B0036 and B0116 exchange referents, B0116 selects that same geometry in B. This is a change in the correct reference, not a change in the canonical part’s type.

The main figures show the actual PNGs, verbatim questions, complete option lists and GT JSON. Identical crops remove empty margins and retain every operand and label. Table 3 exposes the source-record derivation: only the candidate with part type 32250 matches the target, and its current label determines the choice ID. The source identifier and displayed label must be kept distinct. The GT is generated from records, not asserted to have passed human qualification.

`visual-evidence.json` binds every new figure to source captures or exact observation wires, questions, options, golds and crop rectangles. The archived figure provenance remains available for the reused replay. Figure 4 supplies four additional pairs from different sources; selection

Table 2. All 24 source assemblies. Color and Part-type counts are parents, each contributing exactly one primary pair. Shared parts, authors and derived observations do not increase source count.

Source	Name	Band	Parts	Color	Part-type
31028	Sea Plane	D1	53	2	2
31027	Blue Racer	D1	67	2	2
10156	LEGO Truck	D1	111	2	1
omr-42102	Mini CLAAS XERION	D1	129	2	1
omr-42020	Twin-Rotor Helicopter	D1	141	2	1
10036	Pizza To Go	D2	166	2	2
10014	Caboose	D2	170	2	2
omr-31031	Rainforest Animals	D2	215	2	2
omr-31088	Deep Sea Creatures	D2	230	2	1
omr-42004	Mini Backhoe Loader	D2	246	2	2
omr-42061	Telehandler	D2	261	2	2
31009	Small Cottage	D2	270	2	1
5867	Super Speedster	D2	278	2	2
10128	Train Level Crossing	D2	336	2	2
10159	City Airport	D3	592	3	3
10001	Metroliner	D3	848	4	4
omr-10269	Harley-Davidson Fat Boy	D4	1029	4	4
omr-10242	MINI Cooper	D4	1086	4	4
omr-42066	Air Race Jet	D4	1244	5	5
10213	Shuttle Adventure	D4	1254	5	5
omr-10265	Ford Mustang	D4	1357	5	5
10220	Volkswagen T1 Camper Van	D4	1428	5	4
omr-21309	NASA Apollo Saturn V	D4	1845	5	5
omr-42054	CLAAS XERION 5000 TRAC VC	D4	1978	5	5

Table 3. GT derivation for the main Part-type example. Source instance IDs and part types remain constant; the current label of the matching candidate changes. Options C and D refer to B0036 and B0116 respectively, so the required outputs are {"choiceId": "C"} for A and {"choiceId": "D"} for B.

Source instance	Part type	Label in A	Label in B	Role
B0035	32250	B0035	B0035	Target
B0036	32250	B0036	B0116	Match
B0060	32526	B0060	B0060	Distractor
B0116	32449	B0116	B0036	Distractor
B0196	2780	B0196	B0196	Distractor

is illustrative, not based on model outputs or certified answerability.

1.3. Appearance baseline

The legal Color baseline accepts one wire containing only PNG and text/options. It runs in a fresh isolated process with an empty working directory. The evaluator joins IDs and independently reconstructed gold afterward. No private masks, paths, source facts or counterparts are supplied to the predictor.

RGB channels are normalized to $[0, 1]$. Near-white pixels with all channels at least .90 are removed. Pixels whose maximum channel is at most .30 vote Black. Other pixels require saturation at least .35 and hue within 60° of the nearest

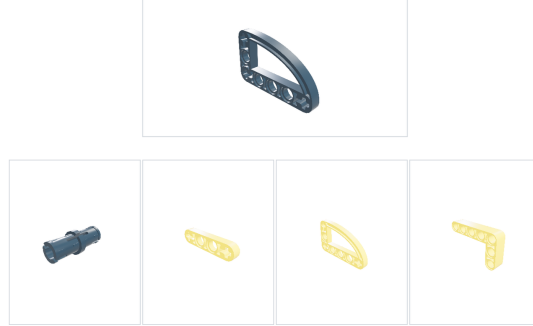
prototype: Red at 0° , Blue at 240° , or Yellow at 60° . Within each category, four-connected components smaller than the greater of 64 pixels and .0005 of image area are removed. The winning category requires .55 of retained support; ties and empty support abstain. Thresholds were locally committed before the corpus run after synthetic tests.

The measured result is 141/146 correct endpoints, five abstentions and 68/73 Both (93.15% micro; 91.875% equal-source). This baseline summarizes rendered appearance across the image; it does not locate the target label or validate multi-object reference following. Per-observation support and failure records are available with the audit.

Position-reference control | Id2-omr-42004-shape-match-1

The upper panel is the target. Which of the four lower candidate panels, read from left to right, shows the same visible part geometry? Ignore color and pose.

A: Leftmost B: Second from left C: Third from left D: Rightmost



GT: {"choiceId":"C"} = Third from left

Figure 1. Auxiliary position-reference question, complete native image, all options and exact GT for OMR 42004. The correct choice is C, third from the left. No labels or GT were added inside the native image. Isolated parts are observations, not a proposed freestanding assembly. The comparison changes layout and visual access together.

2. Part-type Assignment and Priors

Admissible configurations enumerate all four-option permutations and nonmatching swaps with an on-screen anchor and unique source-type match. The 4,824 configurations are ordered by parent, option-label tuple and swap. Constraint-equivalent choices keep the minimum integer hash cost, leaving 2,412 variables. One configuration is selected per parent subject to the stated option, transition and shortcut bounds. The recorded single-thread SciPy 1.17.1/HiGHS solution has objective 1,007,668 with zero relative gap. The frozen assignment, rather than runtime-dependent regeneration, defines the released dataset.

Table 4. A-to-B choice transitions. Rows are A gold, columns B gold. Left: Color (73 pairs); right: Part-type (67 pairs). Balanced semantic colors do not imply balanced option transitions.

A\B	A	B	C	D	A\B	A	B	C	D
A	0	3	3	7	A	0	6	5	5
B	7	0	7	8	B	6	0	6	5
C	4	8	0	7	C	5	6	0	6
D	5	4	10	0	D	6	5	6	0

Table 5. All declared Part-type oracle-A rules: B accuracy (%). A gold is supplied, so B accuracy equals Both. These are privileged construction audits; their intervals are descriptive after assignment.

Oracle-A rule	Micro	Source	Source 95% CI
First alternative	34.33	26.94	[15.62, 38.61]
Last alternative	32.84	32.01	[19.93, 44.86]
Cyclic next	35.82	38.68	[25.00, 53.33]
Cyclic previous	34.33	28.89	[17.92, 40.35]
Cyclic two	29.85	32.43	[19.79, 46.25]
Smallest alternate label	29.85	24.58	[14.58, 34.38]
Largest alternate label	38.81	39.03	[26.04, 52.50]
Nearest label number	29.85	30.42	[18.75, 43.12]
Farthest label number	31.34	29.03	[17.08, 41.53]
Nearest source center	29.85	36.60	[23.61, 50.97]
Farthest source center	35.82	28.75	[17.92, 40.21]
Nearest projected anchor	25.37	30.14	[17.36, 44.38]
Farthest projected anchor	32.84	27.08	[16.25, 38.75]

Rules use first/last alternative, cyclic option position, alternate label-number extrema, distance from target label number, or nearest/farthest source center or projected anchor. Ties use ascending original numeric labels. Spatial rules use evaluator-side geometry. None reads B pixels, and no rule supplies a universal model ceiling. Independent reconstruction checks golds and features against source records and rendering metadata.

3. Human Qualification

3.1. Assignment and raw records

Six packages contain 115, 116, 114, 115, 115 and 119 judgments. Their union covers all 347 observations exactly twice. Reviewers must be distinct people, and each sees at most one observation of any parent, including the panel control. Three-observation parents therefore use all six reviewers. Queues are shuffled and use opaque unit IDs without condition or arm labels.

A raw record contains the reviewer pseudonym, timestamp, observation hash, visual choice, categorical decision, four structured checks, reason, native-size inspection attestation and independence attestation. The four checks concern a unique target, unique visual answer, sufficient visibility and absence of reference contradictions. Decidable judgments require a listed choice and all checks passing. Other decisions require an exclusion reason. The interface hides gold, source data, counterpart, model output and other reviewers' judgments. The coordinator retains unmodified exports.

3.2. Adjudication and denominators

Agreement on decision, choice and checks permits finalization. Any disagreement on these fields requires an additional independent adjudicator. Up to three adjudicator slots preserve parent-disjoint exposure and exclude initial reviewers. The adjudicator sees the single observation and its two judgments, but not gold, another observation of that parent or model outputs. Original records remain intact. A final decidable choice that conflicts with generated gold blocks finalization and requires stimulus revision and renewed review.

The full primary universe is always 280 endpoints and 140 pairs. There are 560 initial primary judgments and 134 auxiliary judgments. Actual collected judgments are currently zero. The runner's existing full-study gate requires final decisions for every planned endpoint. Nondecidable cases remain in full-set scoring; the additional qualified subset requires two decidable endpoints per pair. A new MVP needs a separate frozen subset with valid review lineage, not an override of the full-study gate.

Raw categorical agreement and choice/check agreement use their actual eligible denominators. Pooled two-slot Cohen's kappa is $(p_o - p_e)/(1 - p_e)$ from empirical decision marginals. Because assigned people vary across observations, it is not the reliability of one fixed reviewer pair. Missing judgments or degenerate expected agreement produce undefined kappa. Adjudicated agreement cannot replace the raw measure.

4. Scoring and Dependence

The four B outcomes partition every changed-answer pair: correct new gold, valid old gold, another valid answer, and invalid output. Missing calls are invalid and remain in the fixed denominator. AAcc and Both retain information about A correctness. Conditional adaptation reports the number of A-correct pairs and the sources with at least one eligible pair.

Equal-source estimates average within-source means. The bootstrap resamples whole sources 10,000 times using seed 20260920. Model contrasts use the same pairs. Qualified-minus-full contrasts reuse source weights and recompute means on eligible sources. An empty subset or a draw containing no eligible source is recorded as undefined. No whole-source interval estimates provider randomness across repeats. Boundary outcomes can yield degenerate intervals.

The 24 sources form 18 primary-author and 17 contributor-connected groups. Contributor groups include credited OMR editors. Sensitivity analysis averages within-source outcomes by author, resamples whole author groups and omits each author in turn. Direction changes under omission limit stability claims. The public inventory additionally contains all 276 source-pair part-count and part-set Jaccards; these are inventory similarities, not spatially aligned geometry overlaps.

Table 6. Repeated primary authors. Other primary-author groups contain one source each.

Repeated primary author	Source IDs
Merlijn Wissink	31027, 31028
Philippe Hurbain	42054, 42061, 42066
Robert Paciorek	10036, 10128, 10156
Zoltan Keri	10001, 10159

Table 7. Largest part-inventory overlaps. Count Jaccard uses summed per-type minima divided by summed maxima; set Jaccard ignores counts.

Source A	Source B	Count J.	Set J.
42054	42066	0.450	0.511
42004	42061	0.370	0.354
42004	42020	0.325	0.431
10220	10242	0.322	0.257
10265	10269	0.280	0.320

5. Graph Construction and Completed Algorithm Audits

Graph questions provide explicit undirected edge lists and a node to delete. Repeated edges are deduplicated; surviving

isolates count as components. Independent BFS and union-find agree on all golds. These checks validate the supplied graph operation, not completeness of physical connector detection.

The hypothetical collection has 219 distinct observations: an anchor, a changed partner and an isomorphic invariant partner for each of 73 parents. Strong change contains 12 pairs from eight sources; degree-visible change contains 61 from 23 sources. Each has a corresponding invariance partition. Pair endpoints therefore have 292 appearances across all partitions, but only 219 unique inputs.

Table 9. All strong changed-answer pairs, with matching surviving nodes, edge count, target degree, degree multiset and isolate count. BFS and union-find have Both=1 on each.

Parent task	y_A	y_B	Isolates+1	Both
ld2-10001-graph-removal-4	2	3		0
ld2-10014-graph-removal-1	2	3		0
ld2-10014-graph-removal-2	1	2		0
ld2-10213-graph-removal-1	4	3		0
ld2-31009-graph-removal-2	4	3		0
ld2-5867-graph-removal-1	1	2		0
ld2-omr-10269-graph-removal-1	2	1		0
ld2-omr-10269-graph-removal-2	2	1		0
ld2-omr-10269-graph-removal-4	2	1		0
ld2-omr-21309-graph-removal-2	2	3		0
ld2-omr-21309-graph-removal-4	2	1		0
ld2-omr-42054-graph-removal-4	3	4		0

The strong construction compares a six-cycle and two triangles plus identical isolated nodes. The weaker construction compares a path and a triangle plus isolates. Only the former matches the complete degree multiset. Uniform-feature anonymous 1-WL signatures collide on every strong pair, so a deterministic signature-only predictor cannot answer both different golds correctly. Unique node identifiers, positional features, spectral features and higher-order models lie outside this restriction.

Feature lookups train only on other sources, choose the most frequent answer, break ties toward the smaller integer and abstain on unseen features. Abstention is incorrect. Hypothetical fitting includes the changed endpoints once and excludes invariant duplicates. The isolates-plus-one heuristic predicts the isolate count plus one when any nonisolated node remains; it never traverses components.

Table 10. Completed invariance algorithm audit. Repeating a wrong answer is consistent but fails Both.

Algorithm	Strong / 12	Degree-visible / 61
Node-count lookup	6	31
Edge-count lookup	7	32
Degree-histogram lookup	6	61
Anonymous 1-WL lookup	6	61
Isolates + one	6	61
BFS	12	61
Union-find	12	61

Strong change has six increases and six decreases in gold component count; arm-wise answer histograms need not match. Both is invariant to swapping A and B. Wilson intervals for 0/12 and 12/12 are approximately $[0, .2425]$ and $[\cdot7575, 1]$ under a pair-independence approximation that source reuse violates. The single template and eight-source coverage limit generalization.

5.1. Replay question and action ground truth

The historical task `ld2-omr-42102-source-sequence` asks:

Reveal the supplied source-step window in author order. Actions change visibility only, without simulating insertion trajectories.

Its GT is the ordered action list `[step-1, step-2, step-3, step-4, step-5, step-6]`. Table 11 lists the exact part IDs revealed by each action in Figure 2. This is a supplied-order visibility control, distinct from the primary visual pairs and from physically validated assembly planning.

Table 11. Historical replay GT. Actions reveal the listed IDs at saved poses. Cumulative counts are derived from the recorded source window.

Action	Newly visible IDs	Cumulative parts
step-1	B0001, B0002, B0003	3
step-2	B0004, B0005	5
step-3	B0006, B0007, B0008	8
step-4	B0009, B0010	10
step-5	B0011, B0012	12
step-6	B0013, B0014	14

6. Collection and Publication Contract

The main paper lists 15 proposed models and 306 empty result cells across primary visual, diagnostic, control, graph and mode tables. The model inventory comes from the existing budget snapshot. All result values in `experiments.json` are null and render as empty cells.

Table 8. Completed deterministic algorithm audit. These values are computed algorithm outputs, not planned or measured language-model results. Natural and hypothetical lookups use separate source-held-out fits. Invariance shares anchors with change.

Algorithm	Natural correct / 73	Strong change Both / 12	Degree-visible Both / 61	Strong invariance Both / 12
Node-count lookup	63	0	0	6
Edge-count lookup	44	0	0	7
Degree-histogram lookup	69	0	61	6
Anonymous 1-WL lookup	69	0	61	6
Isolates + one	72	0	61	6
BFS	73	12	61	12
Union-find	73	12	61	12

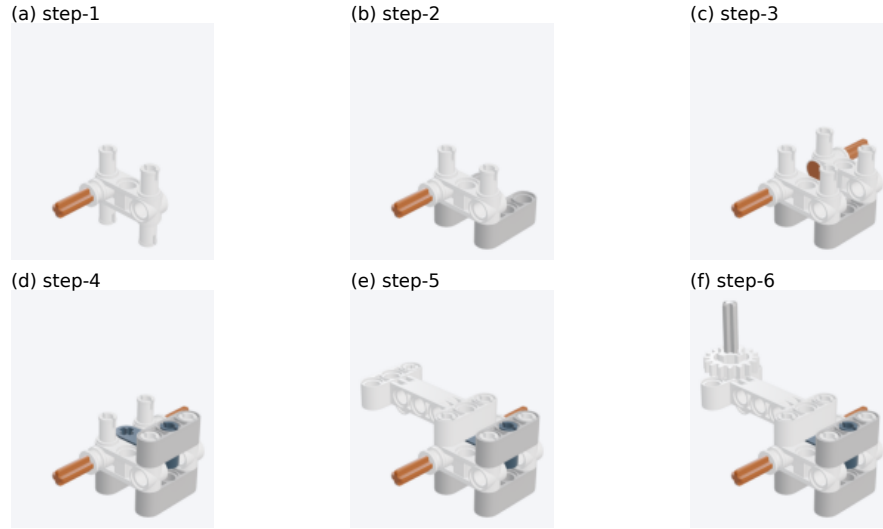


Figure 2. Historical given-order 3D visibility replay for set 42102. Parts appear at their saved source transforms. The supplied sequence is an inspection control, not a discovered model plan; it does not certify an insertion path or gravity stability. Section 5.1 and Table 11 provide the exact question, action IDs and newly visible part IDs.

Null represents uncollected results, not measured zero or an unspecified model name.

Before collection, the new study must freeze selected IDs, exact models and revisions, provider/precision, adapter code, image policy, output and effort settings, repeat count, timeouts, inclusion and failure policies, and qualification lineage. The MVP proposal keeps complete visual pairs and all three observations of each selected graph parent. It requires new adapters and a subset qualification gate; those implementation steps remain open.

The earlier sealed evaluator fixes three models and nine runs. It cannot silently accept the new roster. Its receipt reconstruction, failure denominators and source estimators provide implementation references, while the new study requires its own versioned manifest. No synthetic fixture, dry run, historical response or missing receipt can substitute for a genuine observation in empirical tables. Real response cases must link to exact requests and retained outputs; an

absent failure category remains absent.

Current manuscript files are under `paper/`; code and dataset indexes are under `benchmark/`. Intermediate reviews, guidance and manuscripts are physically archived under `tem/`, with compatibility links for historical paths. Git includes data, image assets, reviewer packages, code and PDFs. Redundant local recovery tarballs and runtimes are excluded; full historical recovery checks require those separate bundles. Hashes establish local consistency, while genuine human independence and provider origin additionally require collection oversight.

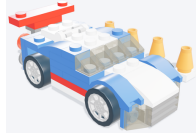
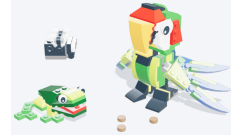
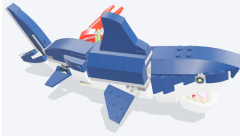
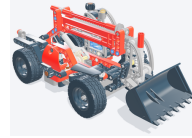
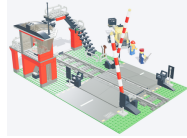
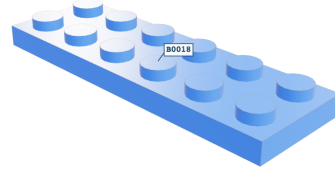
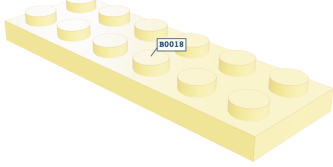
D1 | 31028 | 53 parts
Sea Plane**D1 | 31027 | 67 parts**
Blue Racer**D1 | 10156 | 111 parts**
LEGO Truck**D1 | 42102 | 129 parts**
Mini CLAAS XERION**D1 | 42020 | 141 parts**
Twin-Rotor Helicopter**D2 | 10036 | 166 parts**
Pizza To Go**D2 | 10014 | 170 parts**
Caboose**D2 | 31031 | 215 parts**
Rainforest Animals**D2 | 31088 | 230 parts**
Deep Sea Creatures**D2 | 42004 | 246 parts**
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Telehandler**D2 | 31009 | 270 parts**
Small Cottage**D2 | 5867 | 278 parts**
Super Speedster**D2 | 10128 | 336 parts**
Train Level Crossing**D3 | 10159 | 592 parts**
City Airport**D3 | 10001 | 848 parts**
Metroliner**D4 | 10269 | 1,029 parts**
Harley-Davidson Fat Boy**D4 | 10242 | 1,086 parts**
MINI Cooper**D4 | 42066 | 1,244 parts**
Air Race Jet**D4 | 10213 | 1,254 parts**
Shuttle Adventure**D4 | 10265 | 1,357 parts**
Ford Mustang**D4 | 10220 | 1,428 parts**
Volkswagen T1 Camper Van**D4 | 21309 | 1,845 parts**
NASA Apollo Saturn V**D4 | 42054 | 1,978 parts**
CLAAS XERION 5000 TRAC VC

Figure 3. All 24 source assemblies, retaining original LDraw geometry and saved poses. IDs and counts are read from the source catalog; the images are hash-verified original browser captures. Scale bands describe source size. These views are for resource inspection and are not substituted for frozen primary model inputs.

Color | Id2-31028-color-1

What color is the object currently marked B0018 in this image?

A: Black B: Red C: Yellow D: Blue



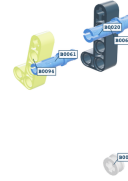
A GT: {"choiceId":"C"} = Yellow

B GT: {"choiceId":"D"} = Blue

Part-type | Id2-omr-42102-shape-match-1

Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B0094? Ignore color and pose. Answer from this image, not from historical source identities.

A: B0003 B: B0061 C: B0020 D: B0065



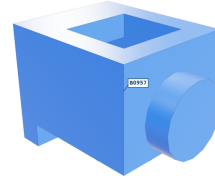
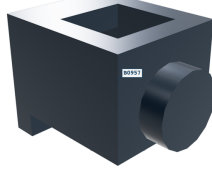
A GT: {"choiceId":"B"} = B0061

B GT: {"choiceId":"C"} = B0020

Color | Id2-10213-color-1

What color is the object currently marked B0957 in this image?

A: Red B: Yellow C: Blue D: Black



A GT: {"choiceId":"D"} = Black

B GT: {"choiceId":"C"} = Blue

Part-type | Id2-omr-42054-shape-match-2

Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B1016? Ignore color and pose. Answer from this image, not from historical source identities.

A: B0191 B: B1766 C: B1357 D: B0439



A GT: {"choiceId":"D"} = B0439

B GT: {"choiceId":"C"} = B1357

Figure 4. Four further real paired examples with complete question, options and GT: two Color and two Part-type parents from four sources. Within each pair only shared empty margins are cropped; all operands remain in their source poses. Green text is evaluator GT outside model input. No generated model responses or human judgments are shown.